

k-means-clustering

```
[22]: import pandas as pd
import numpy as np
import seaborn
import matplotlib.pyplot as plt
```

```
[12]: df= pd.read_csv('Mall_Customers.csv')
```

```
[13]: df
```

```
[13]:
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
..	...	...	...	...	...
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

[200 rows x 5 columns]

```
[15]: df.rename (columns = {'Gender': 'gender', 'Age': 'age', 'Annual Income (k$)': 'income', 'Spending Score (1-100)': 'score' },inplace = True)
```

```
[16]: df
```

```
[16]:
```

	CustomerID	gender	age	income	score
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40
..	...	...	...	...	...
195	196	Female	35	120	79

196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

[200 rows x 5 columns]

```
[9]: df.shape
```

```
[9]: (200, 5)
```

```
[11]: df.isnull().values.any()
```

```
[11]: False
```

```
[12]: df.describe()
```

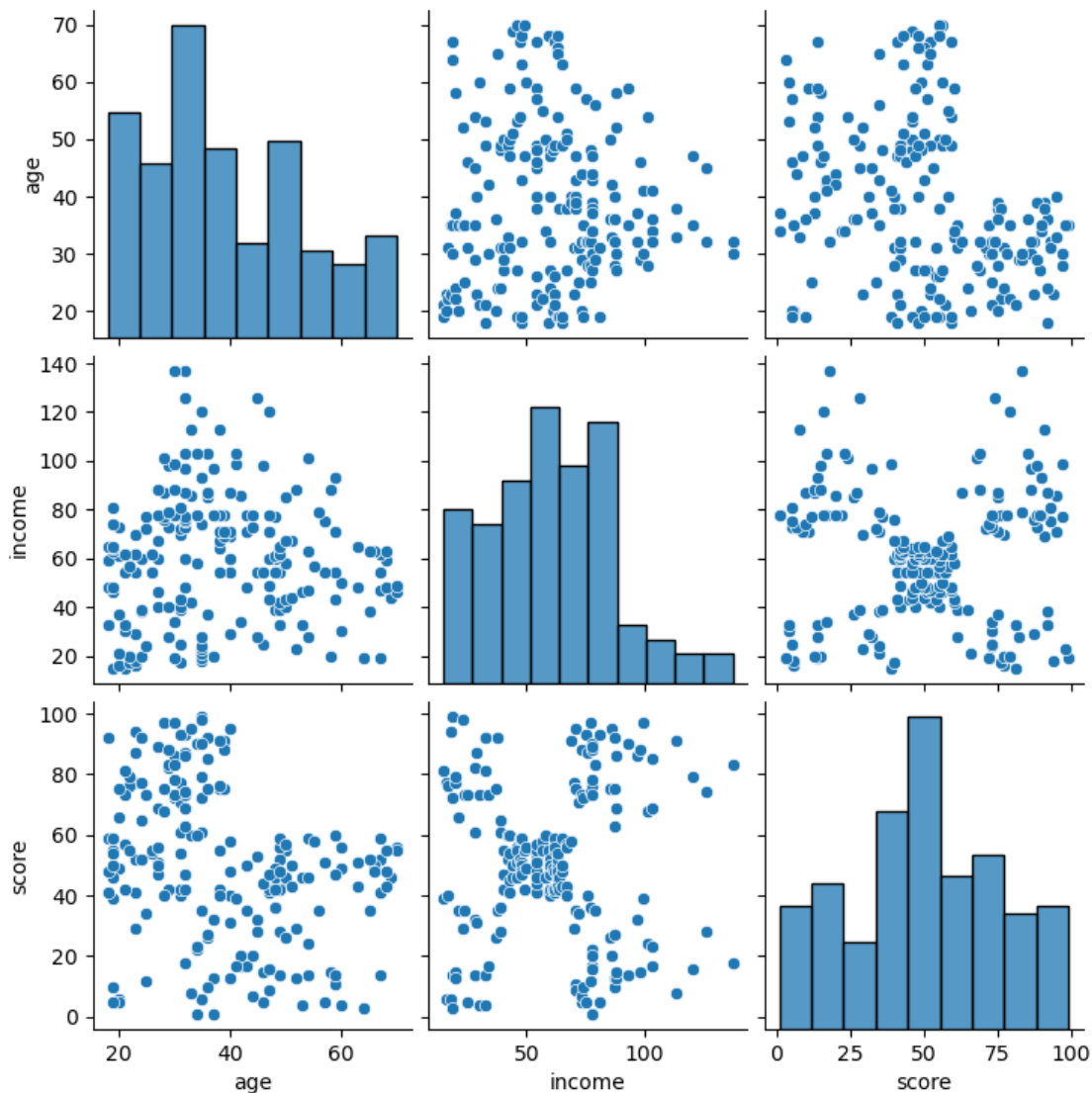
```
[12]:
```

	CustomerID	age	income	score
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

```
[15]: seaborn.pairplot (df[['age', 'income', 'score']])
```

```
C:\ProgramData\anaconda3\Lib\site-packages\seaborn\axisgrid.py:118: UserWarning:
The figure layout has changed to tight
    self._figure.tight_layout(*args, **kwargs)
```

```
[15]: <seaborn.axisgrid.PairGrid at 0x1f152677b50>
```



```
[16]: import sklearn.cluster as cluster
```

```
kmeans = cluster.KMeans(n_clusters = 5)
```

```
[24]: kmeans = kmeans.fit(df[['score', 'income']])
```

```
C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1412:
FutureWarning: The default value of `n_init` will change from 10 to 'auto' in
1.4. Set the value of `n_init` explicitly to suppress the warning
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C:\ProgramData\anaconda3\Lib\site-packages\sklearn\cluster\_kmeans.py:1436:
UserWarning: KMeans is known to have a memory leak on Windows with MKL, when
there are less chunks than available threads. You can avoid it by setting the
```

```
environment variable OMP_NUM_THREADS=1.
warnings.warn(
```

```
[25]: #Finding out the centroids
```

```
kmeans.cluster_centers_
```

```
[25]: array([[82.12820513, 86.53846154],
            [20.91304348, 26.30434783],
            [49.51851852, 55.2962963 ],
            [17.11428571, 88.2         ],
            [79.36363636, 25.72727273]])
```

```
[26]: kmeans = kmeans.fit(df[['income', 'score']])
```

```
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```

```
[27]: kmeans.cluster_centers_
```

```
[27]: array([[25.72727273, 79.36363636],
            [55.2962963 , 49.51851852],
            [86.53846154, 82.12820513],
            [88.2         , 17.11428571],
            [26.30434783, 20.91304348]])
```

```
[28]: df['income_clusters'] = kmeans.labels_
```

```
[29]: df
```

```
[29]:
```

	CustomerID	gender	age	income	score	income_clusters
0	1	Male	19	15	39	4
1	2	Male	21	15	81	0
2	3	Female	20	16	6	4
3	4	Female	23	16	77	0
4	5	Female	31	17	40	4
..	...	...	...	...	...	...
195	196	Female	35	120	79	2
196	197	Female	45	126	28	3
197	198	Male	32	126	74	2
198	199	Male	32	137	18	3

```
199          200      Male    30      137      83          2
```

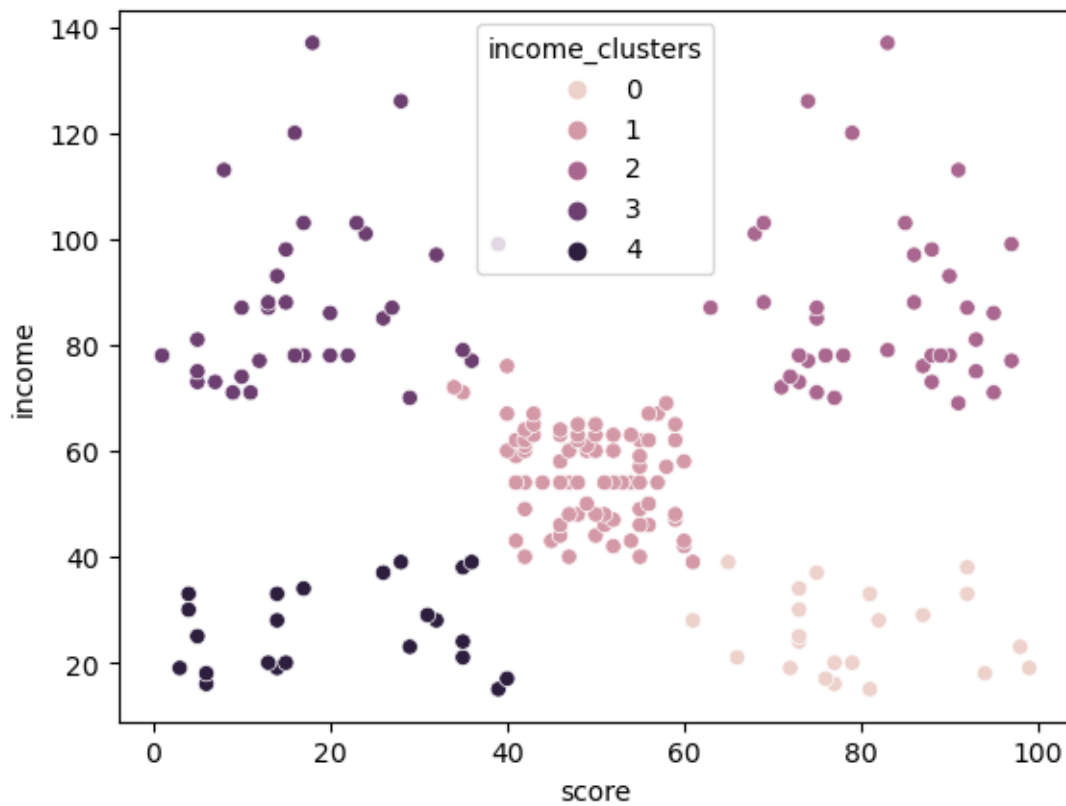
```
[200 rows x 6 columns]
```

```
[30]: # counting in which cluster how many values
df['income_clusters'].value_counts()
```

```
[30]: income_clusters
1      81
2      39
3      35
4      23
0      22
Name: count, dtype: int64
```

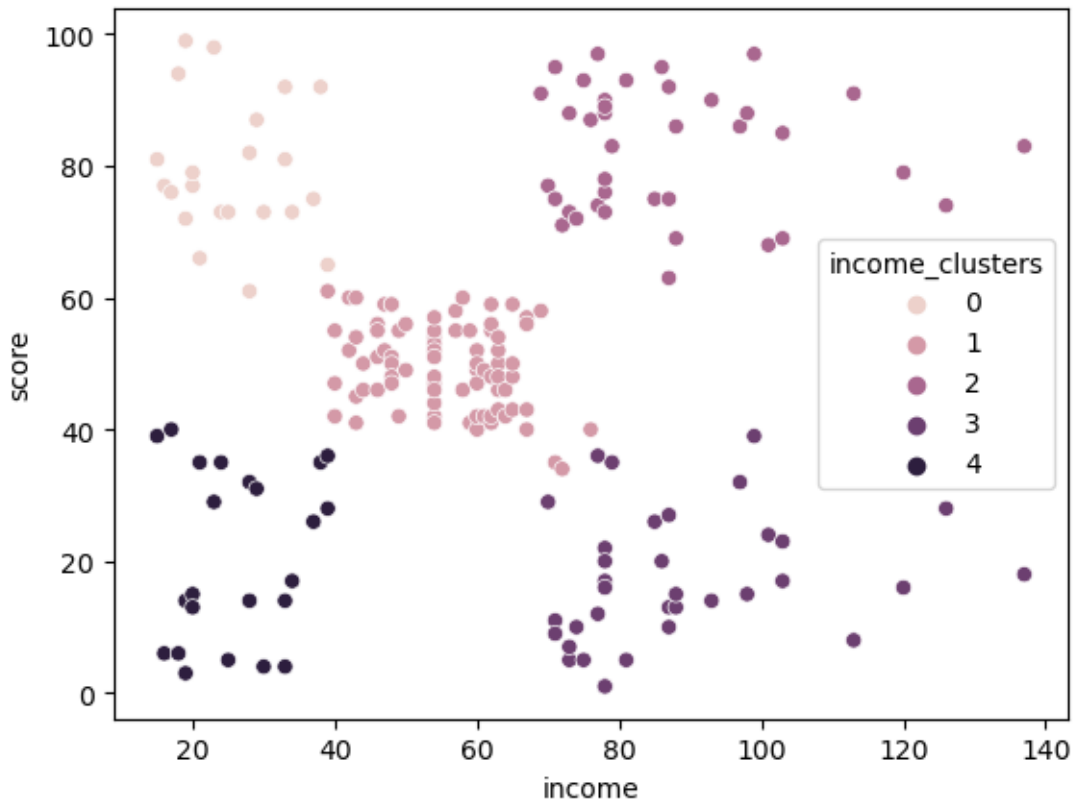
```
[31]: seaborn.scatterplot(x= 'score', y = 'income', hue= 'income_clusters', data= df)
```

```
[31]: <Axes: xlabel='score', ylabel='income'>
```



```
[32]: seaborn.scatterplot(x= 'income', y = 'score', hue= 'income_clusters', data= df)
```

```
[32]: <Axes: xlabel='income', ylabel='score'>
```



```
[33]: #trying with 2 clusters
kmeans = cluster.KMeans(n_clusters = 2)
```

```
[34]: kmeans = kmeans.fit(df[['age', 'score']])
```

```
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there are less chunks than available threads. You can avoid it by setting the
environment variable OMP_NUM_THREADS=1.
  warnings.warn(
```

```
[35]: kmeans.cluster_centers_
```

```
[35]: array([[28.95294118, 73.62352941],
          [46.16521739, 32.88695652]])
```

```
[36]: df['age_clusters'] = kmeans.labels_
```

```
[37]: df
```

```
[37]:
```

	CustomerID	gender	age	income	score	income_clusters	age_clusters
0	1	Male	19	15	39	4	1
1	2	Male	21	15	81	0	0
2	3	Female	20	16	6	4	1
3	4	Female	23	16	77	0	0
4	5	Female	31	17	40	4	1
..	...	...	...	...	...	...	...
195	196	Female	35	120	79	2	0
196	197	Female	45	126	28	3	1
197	198	Male	32	126	74	2	0
198	199	Male	32	137	18	3	1
199	200	Male	30	137	83	2	0

[200 rows x 7 columns]

```
[38]: df['age_clusters'].value_counts()
```

```
[38]: age_clusters
1      115
0       85
Name: count, dtype: int64
```

```
[39]: df
```

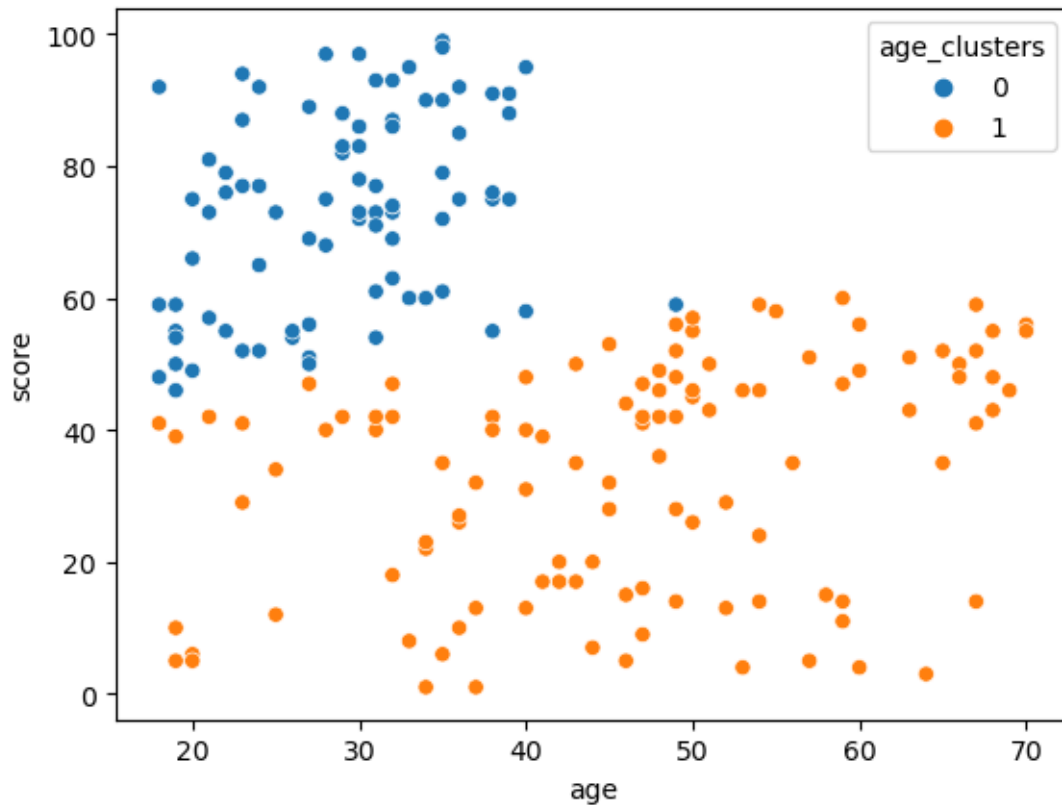
```
[39]:
```

	CustomerID	gender	age	income	score	income_clusters	age_clusters
0	1	Male	19	15	39	4	1
1	2	Male	21	15	81	0	0
2	3	Female	20	16	6	4	1
3	4	Female	23	16	77	0	0
4	5	Female	31	17	40	4	1
..	...	...	...	...	...	...	...
195	196	Female	35	120	79	2	0
196	197	Female	45	126	28	3	1
197	198	Male	32	126	74	2	0
198	199	Male	32	137	18	3	1
199	200	Male	30	137	83	2	0

[200 rows x 7 columns]

```
[40]: seaborn.scatterplot(x= 'age', y = 'score', hue= 'age_clusters', data= df)
```

```
[40]: <Axes: xlabel='age', ylabel='score'>
```



1 Applying Elbow methods

```
[3]: #from sklearn.cluster import kMeans
from sklearn.cluster import KMeans
```

```
[6]: # to calculate 12 clusters
```

```
k_range = range(1,12)
wcscs = []
```

```
[20]: # implementing for loop for 12 clusters to find out wcscs value for each cluster
# adding wcscs values of clusters in wcscs array
```

```
for k in k_range:
    km= KMeans(n_clusters =k)
    km.fit(df[['income','score']])
    wcscs.append (km.inertia_)
```

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```
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```

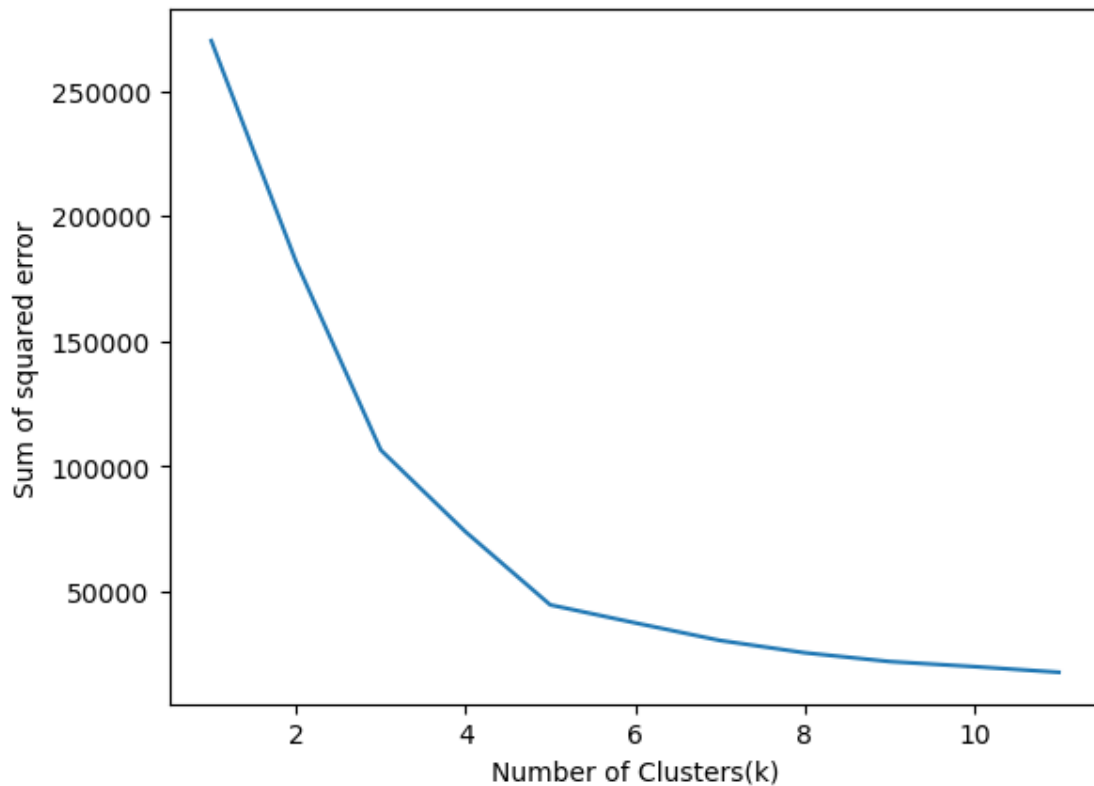
2 showing the values of wcss

```
[21]: wcss
```

```
[21]: [269981.28,  
      181665.82312925172,  
      106348.37306211122,  
      73679.78903948836,  
      44448.4554479337,  
      37265.86520484346,  
      30241.34361793658,  
      25315.541822712163,  
      21809.92756261518,  
      19770.06950274374,  
      17511.7418692661]
```

```
[23]: plt.xlabel ('Number of Clusters(k)')  
      plt.ylabel ('Sum of squared error')  
      plt.plot (k_range , wcss)
```

```
[23]: [<matplotlib.lines.Line2D at 0x2098852b0d0>]
```



3 Optimal number of k is 5